QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

FranchiseIQ

AI Decision Intelligence Dashboard for Retail Franchise Management

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Document Control

Field	Detail
System Under Test (SUT)	FranchiseIQ	UMHackathon 2026	AI Decision Intelligence Dashboard
Team Repository URL	https://github.com/Ywintersss/Retailers
Project Board URL	https://www.notion.so/34d52308a3f480f395cec3a368cefa96?v=12096756369240658d7d9a4fbbd1ec87&source=copy_link
Live Deployment URL	https://retailers.webhop.me/ (mock-data mode available without backend)
Document Version	v1.0	Preliminary Round Submission

Objective

The primary objective of this Quality Assurance Testing Documentation is to ensure that FranchiseIQ reliably handles real-time POS data ingestion, AI-driven pricing recalculations, inventory forecasting, and sentiment scoring under both normal and concurrent load conditions. The document establishes the testing strategy, defines risk-mitigation approaches for the three highest-priority technical risks, and provides traceable test cases for each core feature. Automated CI/CD checkpoints are defined to prevent regression from being introduced during active development, and every AI inference boundary is tested explicitly to ensure the Z AI GLM integration remains deterministic, schema-valid, and within acceptable business constraints throughout the hackathon period.

1. Scope & Requirements Traceability

This section aligns each testing activity back to a specific user requirement through a Requirements Traceability approach. The goal is to ensure that every feature the team has committed to building has at least one corresponding test case, and that no unplanned feature creep is validated without a defined acceptance criterion. For FranchiseIQ, the scope of testing is bounded by the four AI Decision Node features and the Kafka ingestion pipeline, as these are the components that carry the highest technical and business risk.

1.1. In-Scope Core Features

The following features are confirmed in-scope for testing during the UMHackathon 2026 preliminary round.

- AI Dynamic Pricing. This covers the weight-based trade-off optimisation, the debounced slider interaction on the frontend, the backend weight validation logic (ensuring the three weights sum to exactly 100%), and the GLM-generated per-SKU pricing recommendation with the PriceValidator sanity bounds applied.
- Sentiment Tracking and Scorecard. This covers the NLP analysis of POS reviews and equipment alert signals, the aggregation of scores into a single 1-to-5 store health rating, the GLM-generated plain-language summary, and the critical alert detection logic for food safety clusters.
- Kafka POS Ingestion. This covers the high-throughput ingest of POS transaction events through the POST /events/pos/ingest endpoint, the Kafka consumer's ability to process events asynchronously without blocking the REST API layer, and the correct display of events on the Real-time Event Stream Feed within an acceptable delay.
- Inventory Forecasting. This covers the 7-day demand prediction per SKU, the GLM's cross-referencing of weather signals, school calendar data, and historical POS history, and the confidence scoring and stockout risk classification returned in the response.
- Actionable Insights Generation. This covers the synthesis of signals from the three other feature modules into a priority-ranked recommendation list, with each insight carrying a plain-language action, an estimated RM impact, and a confidence score.

1.2. Out-of-Scope

The following items are explicitly excluded from the testing scope for the preliminary round, as they fall outside the MVP boundaries defined in the System Analysis Documentation.

- Multi-user real-time collaboration or shared session features between concurrent managers.
- Production-grade payment gateway integration or any financial transaction processing.
- Native mobile application testing. The current version is web-only and tested exclusively on desktop browsers.
- Profile management, account settings, and user authentication flows beyond what is needed to load the dashboard.
- Multi-store aggregate dashboards and cross-tenant data access, which are reserved for the post-hackathon roadmap.

2. Risk Assessment & Mitigation Strategy

Quality assurance risks are anticipatory in nature. The following table identifies the highest-priority technical risks associated with FranchiseIQ's architecture, evaluated using the 5x5 Risk Assessment Matrix where Risk Score = Likelihood x Severity. Risks with a score of 11 or above require active mitigation strategies and thorough testing coverage before the system can be considered stable.

Likelihood (1-5)	Severity (1-5)
1 | Rare	1 | Negligible Impact
2 | Unlikely	2 | Minor Impact
3 | Possible	3 | Moderate Impact
4 | Likely	4 | Major Impact
5 | Almost Certain	5 | Critical System Failure

Technical Risk	Likelihood (1-5)	Severity (1-5)	Risk Score	Mitigation Strategy	Testing Approach
AI Hallucination in Pricing Output	**2**	**5**	**10 (Med)**	Implement a backend PriceValidator service that hard-codes sanity bounds of +/- 20% around the current SKU price. Any GLM output exceeding these bounds is clamped and flagged in the response.	Unit tests on the PriceValidator service using boundary-value inputs, including prices exactly at the 20% limit, 1 unit above it, and prices that are clearly nonsensical.
Kafka Topic Backlog Under Load	**3**	**4**	**12 (High)**	Kafka consumer operates fully asynchronously and is decoupled from the REST API layer. Consumer group lag is monitored via the Kafka AdminClient API, with an alert threshold of 10,000 unprocessed messages.	Load testing using a mock Kafka producer that fires 10,000+ POS events within 1 second. The test validates that all events are consumed without errors and that the REST API response time is unaffected during the ingestion burst.
Invalid JSON Schema Returned by GLM	**3**	**5**	**15 (High)**	LangChain4j Structured Output Parsers enforce a strict DTO schema on every GLM response. If the first response fails validation, the system retries once with a schema	Integration tests using deliberately malformed JSON mocks injected in place of the GLM API response. Tests validate that the parser retries correctly, and that the fallback mechanism returns the cached result

Technical Risk	Likelihood (1-5)	Severity (1-5)	Risk Score	Mitigation Strategy	Testing Approach
				correction prompt. If the retry also fails, it falls back to the cached PostgreSQL result.	with a CACHED_RESULT flag.
GLM API Rate Limit Exceeded	**2**	**4**	**8 (Med)**	React Query client-side caching holds AI responses for 24 hours on static data such as sentiment baselines. Debounced slider inputs (1-second delay) prevent redundant concurrent calls to the GLM API.	Simulate rapid consecutive slider adjustments (10 within 2 seconds) and validate that only 1 GLM call is triggered due to the debounce logic. Verify the cached response is served for repeated identical queries.
Frontend Weight Auto-Balance Bypass	**2**	**3**	**6 (Med)**	The backend validates that the three pricing trade-off weights (Cost Saving, Brand Presence, Revenue Maximisation) sum to exactly 100 on every POST request. Any non-100 sum results in a 400 Bad Request response.	Negative test case TC-02 sends a manually crafted request payload where the weights sum to 150%. The test validates that the backend rejects it with 400, and that the frontend auto-balance logic prevents the user from reaching this state under normal interaction.

Risk Score	Risk Level	Recommended Action
1 to 5	**Low**	Monitor only. Acceptable risks that do not require active intervention during the hackathon period.
6 to 10	**Medium**	Mitigate with targeted implementation controls and include at least one corresponding test case.
11 to 15	**High**	Must have an active mitigation strategy in place and thorough test coverage before any deployment.
16 to 25	**Critical**	Highest priority. Requires extensive testing, approval from all team members, and a documented rollback plan.

3. Test Environment & Execution Strategy

This section describes where testing takes place, how test data is managed, and the rules that govern each testing phase. FranchiseIQ uses a four-tier testing strategy that mirrors the development lifecycle, ensuring that individual units of logic are validated before integration points are tested, and that regression is checked automatically on every code merge.

3.1. Unit Tests

- Scope. Backend Java domain logic, including the PriceValidator service, the DTO Flattening service, the weight validation logic in the PricingController, and all domain entity methods. Frontend React utility functions, including the slider auto-balance logic in the DynamicPricingPanel component and the TanStack Query hook configurations in useApiQueries.js.
- Execution. Backend unit tests are written using JUnit 5 and run automatically on every commit to any branch via the CI pipeline. Frontend unit tests are written using Vitest and are also triggered on every push. Both suites run in under 30 seconds to avoid blocking developer flow.
- Isolation. Mockito is used on the backend to mock all external dependencies, including the PostgreSQL database, the Kafka consumer, and the LangChain4j GLM client. This ensures that the logic of each class is tested in complete isolation from infrastructure concerns. On the frontend, Vi Mocks replace all TanStack Query hooks and API client calls so that component logic can be tested without a live backend.
- Pass Condition. Every unit test must pass for all happy cases, all negative cases (e.g., a null SKU identifier, a weight payload summing to 101%, and a GLM client that throws a timeout exception), and all relevant boundary values. A test that returns a generic 500 error where a descriptive validation message is expected is treated as a failure.

3.2. Integration Tests

- Scope. Controller-to-Use-Case communication, specifically the flow from the PricingController through the PricingOptimizationUseCase and into the mocked LangChain4j GLM client. This validates that the DTO flattening, prompt construction, response parsing, and schema validation steps all interact correctly as a chain.
- Execution. Integration tests are run after the unit test suite passes. They use the @WebMvcTest annotation with MockMvc on the backend, loading only the web layer and mocking the service layer. This approach is faster than a full SpringBootTest but still validates the HTTP contract, serialisation, and error-handling behaviour.
- Workflow. The physical Z AI GLM API is never called during integration testing. Instead, the LangChain4j client is replaced with a mock that returns a pre-defined JSON fixture. Separate test fixtures cover the happy-case valid JSON response, a response that fails schema validation, and a response that exceeds the pricing sanity bounds. This allows the team to test the full parsing and fallback pipeline without depending on an external API.
- Pass Condition. Integration tests pass when the integrated components produce the exact expected HTTP status codes, response bodies matching the documented DTO schemas, and the correct database interactions (verified through an H2 in-memory database seeded with test data).

3.3. Test Environment (CI/CD Practice)

- Local Testing. Developers run unit and integration tests manually on localhost before pushing to any branch. The frontend runs at http://localhost:5173 with mock data active when the backend is unavailable, allowing frontend testing to proceed independently.
- Staging and CI. GitHub Actions triggers the full unit and integration test suite on every push to any branch. Test results are reported in the pull request checks panel. A failing test suite blocks the pull request from being reviewed or merged.
- CI/CD Automated Pipeline. For pull requests targeting the main branch, the pipeline runs unit tests, integration tests, and a static code quality check. The deployment step is gated behind all three checks passing simultaneously. Branch-specific pushes run unit tests only to keep the feedback loop fast.

3.4. Regression Testing & Pass/Fail Rules

- Execution Phase. Regression tests run across all five in-scope features each time a new branch is merged to main. This is automated through the GitHub Actions pipeline and uses the full JUnit 5 and Vitest suites plus a set of designated regression scenarios.
- Pass/Fail Condition. A test case is considered passing only when the actual output exactly matches the expected output defined in its specification, including the HTTP status code, the response body structure, and all business constraint validations. Any mismatch, even a minor one such as an unexpected null field or a missing CACHED_RESULT flag on a fallback response, is recorded in the Defect Log and must be addressed before the next deployment.
- Continuation Rule. The pipeline enforces a sequential gate: unit tests must all pass before integration tests are run, and integration tests must meet their threshold before regression tests are executed. Skipping a failed gate is not permitted.

3.5. Test Data Strategy

- Manual. Five mock store records were created in the local H2 test database covering a Tier 1 high-traffic outlet (store-kl-001), a Tier 2 mid-traffic outlet, a store with a simulated broken freezer alert active, a store with all KPIs in healthy ranges, and a store with extreme sales history data to test the 30-day rolling window truncation logic.
- Automated. SQL seed scripts populate the test database with 90 days of POS transaction history per SKU before each integration test run, ensuring consistent and reproducible data conditions for all pricing and forecasting test scenarios.
- AI Mock Fixtures. Four JSON fixture files represent the GLM response scenarios: a valid pricing recommendation, a valid sentiment analysis result, a malformed JSON response to trigger the retry logic, and a valid response where one SKU price exceeds the 20% sanity bound.

3.6. Passing Rate Thresholds

- Critical Path (AI Pricing Recalculation, POS Ingestion): 100% pass rate required with no exceptions. A failure in either of these flows blocks all further pipeline stages and requires an immediate fix before any code can be merged to main.
- Overall Application Layer: A minimum of 90% of all test cases executed across the core application layer must pass. This threshold applies to the combined JUnit 5 and Vitest results for each CI run.

- AI Output Pass Rate: A minimum of 80% of the documented prompt/response test pairs must return structurally valid JSON that passes schema validation, for the system to be considered stable for a deployment candidate.

4. CI/CD Release Thresholds & Automation Gates

This section defines the specific metrics and thresholds that the CI/CD pipeline uses as success criteria. If a threshold is not met, the pipeline blocks forward transition automatically. These gates are the team's primary defence against regression and unstable AI outputs reaching a reviewable state.

4.1. Integration Thresholds (Merging to Main Branch)

The main branch is the stable source of truth for the project. Every pull request must pass all four of the following checks before a merge is permitted.

Check	Requirement	Current Project Status	Gate Result
Automatic Build	Zero build errors	0 errors	**Passed**
Unit Tests (JUnit 5 + Vitest)	100% passing rate	100%	**Passed**
Code Quality (ESLint + Checkstyle)	Zero linting errors	3 warnings, 0 errors	**Passed**
Test Coverage (Application Layer)	Minimum 90% coverage	91.4%	**Passed**

4.2. Deployment Thresholds (Pushing to Production / Demo)

The following thresholds must be satisfied before the build is approved for deployment to the demo environment used during the hackathon presentation.

Check	Requirement	Current Project Status	Gate Result
Regression Test Suite	Minimum 90% pass rate	92.3%	**Passed**
AI Output Pass Rate	Minimum 80% of prompt pairs	85% (17 of 20 pairs)	**Passed**
Critical Bugs (P0/P1)	Zero P0 or P1 bugs open	0 open	**Passed**
API Performance	p95 response time under 5 seconds	3.2s at p95	**Passed**
Security: No Exposed Secrets	No API keys or credentials in source code	Clean (GitLeaks scan)	**Passed**
Kafka Consumer Lag	Consumer lag below 10,000 messages	Max 420 messages in load test	**Passed**

5. Test Case Specifications

This section contains the full set of test cases covering the required categories: one happy case that validates an end-to-end user journey, one specific negative or edge case, and two non-functional requirement tests covering performance and load. An additional set of AI-specific test cases is documented separately in Section 6. All test cases include the mapped feature, a description, step-by-step test steps, the expected result, the actual result observed during testing, and the current pass/fail status.

TC-01 | Happy Case (Entire Flow): Dynamic Pricing Recalculation

Field	Detail
Test Type	Happy Case (Entire End-to-End Flow)
Mapped Feature	Dynamic Pricing	Human-in-the-Loop Weight Recalculation
Description	This test validates the complete pricing recalculation journey from the manager's slider adjustment on the React frontend, through the Spring Boot backend, through the Z AI GLM inference, and back to the updated price list displayed in the UI. All modules must communicate correctly for this test to pass.

Test Steps

- Open the FranchiseIQ dashboard at http://localhost:5173 and ensure the DynamicPricingPanel is visible on the main view.
- Verify the current trade-off weights are at their default state: Cost Saving at 40%, Brand Presence at 35%, and Revenue Maximisation at 25%.
- Drag the Revenue Maximisation slider gradually to 80%. Observe that the auto-balance logic redistributes the remaining 20% proportionally across Cost Saving and Brand Presence without the total exceeding 100%.
- Pause interaction for 1.1 seconds to allow the debounce timer to expire. Observe that the RECALCULATING badge appears on the pricing panel immediately after the debounce period.
- Wait for the AI inference to complete. Observe the panel transition from the loading state to the updated price list.
- Verify the response contains a recommendations array with at least one SKU entry, where the recommendedPrice for each item reflects an upward adjustment consistent with the revenue-maximisation priority.
- Verify the projectedRevenue block shows a withOptimization value higher than withoutOptimization, and that the uplift percentage is greater than 0.
- Click the Explain Decision panel on any SKU recommendation and verify that it displays the data variables used, the GLM model version (Z.AI GLM v3.2), and the reasoning chain in plain language.

Expected Result	The RECALCULATING badge appears within 1.1 seconds of the last slider movement. The GLM returns a valid pricing recommendation JSON within 5 seconds at p95. The UI updates the per-SKU prices, the projected revenue uplift is displayed with a positive RM value, and the Explain Decision panel

	is populated with non-empty reasoning text. The Kafka Event Stream Feed continues updating in the background, confirming that the AI inference did not block the ingestion pipeline.
Actual Result	The RECALCULATING badge appeared after 1.05 seconds. The GLM returned a valid response in 2.8 seconds. All 6 active SKUs received updated recommended prices with the Brown Sugar Boba item increasing from RM 8.90 to RM 9.50 under the revenue-maximisation strategy. The projected uplift was displayed as +RM 1,750 (14.0%). The Explain Decision panel loaded correctly with 4 data variables, including the weather forecast and school calendar signals. The Event Stream Feed continued receiving events throughout.
Status	**Passed**

TC-02 | Specific Case (Negative): Invalid Weight Sum Bypass Attempt

Field	Detail
Test Type	Specific Case (Negative / Edge Test)
Mapped Feature	Dynamic Pricing	Backend Weight Validation and Frontend Auto-Balance
Description	This test verifies that neither the frontend nor the backend will allow a pricing recalculation request to proceed when the three trade-off weights do not sum to exactly 100. The system must handle this gracefully at both layers without triggering an unhandled exception or a blank screen.

Test Steps

- Using the Postman API client, send a direct POST request to PUT /ai/pricing/weights/store-kl-001 with the following manually crafted payload: { "costSaving": 60, "brandPresence": 50, "revenueMaximization": 40 }, which sums to 150.
- Observe the HTTP response status code and response body returned by the Spring Boot backend.
- In a second sub-test, open the FranchiseIQ React dashboard and attempt to interact with the sliders rapidly to produce a state where the sum momentarily exceeds 100.
- Observe whether the frontend auto-balance logic intervenes before any API call is made.
- Verify in the browser network panel that no POST request was sent while the weight sum was invalid.

Expected Result	The backend returns HTTP 400 Bad Request with a descriptive error body such as { "error": "Weight sum must equal 100. Received: 150" } for the direct API call. On the frontend, the auto-balance logic prevents the sum from ever exceeding 100 during slider interaction, so no invalid API call is ever dispatched to the backend.
Actual Result	The direct Postman call returned HTTP 400 with the error message "Trade-off weights must sum to exactly 100. Received sum: 150." The frontend slider testing confirmed that the auto-balance logic redistributed weight in real-time on every slider movement, and no network request with an invalid sum was captured in the browser developer tools network panel.
Status	**Passed**

TC-03 | NFR (Performance): AI Response Latency Under Load

Field	Detail
Test Type	NFR Test	Performance and Latency
Mapped Feature	AI Dynamic Pricing	Non-Functional Requirement: Token Latency (p95 < 5 seconds)
Description	This test verifies that the AI pricing recalculation endpoint maintains an acceptable p95 response time of under 5 seconds under simulated concurrent load, and that the RECALCULATING loading state is visible on the frontend throughout the entire wait period.

Test Steps

- Open Postman and configure a Collection Runner targeting the POST endpoint PUT /ai/pricing/weights/store-kl-001 with a valid weight payload.
- Execute 20 consecutive requests to the endpoint with no delay between iterations to simulate a rapid sequence of slider adjustments under realistic demo conditions.
- Record the response time for each of the 20 requests.
- Calculate the p95 response time by sorting the 20 values and taking the value at the 19th position.
- In a parallel browser session, perform the same slider interaction 5 times and visually confirm that the RECALCULATING badge appears immediately after the debounce period expires on each attempt.
- Verify that no request in the 20-request sequence returns a null recommendations array or an empty JSON object.

Expected Result	The p95 response time across all 20 requests is under 5 seconds. The RECALCULATING badge is visible on all 5 browser-session test runs for the full duration of the AI inference. No request returns an empty recommendations array or a null response body.
Actual Result	The 20 requests recorded response times ranging from 1.9 seconds to 4.1 seconds. The p95 value, taken at the 19th result when sorted, was 3.8 seconds, which is within the 5-second threshold. The RECALCULATING badge was confirmed visible on all 5 browser-session runs. All 20 requests returned a valid recommendations array with at least 4 SKU entries each.
Status	**Passed	p95 = 3.8s (threshold: < 5s)**

TC-04 | NFR (Load): Kafka POS Event Throughput

Field	Detail
Test Type	NFR Test	Load and Throughput
Mapped Feature	Kafka POS Ingestion	Non-Functional Requirement: Scalability
Description	This test verifies that the Kafka consumer pipeline can process 100 POS transaction events within a 1-second burst, that the backend returns 202 Accepted immediately for all events without blocking, and that all events appear in the Real-time Event Stream Feed on the dashboard within an acceptable delay.

Test Steps

- Start the mock Kafka producer script configured to publish 100 POS transaction events to the pos.transactions topic within a 1-second window.
- Each event follows the POST /events/pos/ingest request schema, with a valid storeId, terminalId, a Brown Sugar Boba line item, and a sequential timestamp.
- Observe the HTTP response returned by the backend for each event. Record the status codes and any error responses.
- Check the consumer group lag metric via the Kafka AdminClient dashboard immediately after the burst completes.
- Open the FranchiseIQ dashboard and observe the Real-time Event Stream Feed panel. Record the time from the first event being published to the first event appearing in the feed ticker.
- Verify that no events are duplicated or dropped by comparing the event count in the feed against the 100 events published.

Expected Result	The backend returns HTTP 202 Accepted for all 100 events. Consumer group lag does not exceed 10,000 messages at any point. All 100 events appear in the Event Stream Feed within 3 seconds of publication. No events are duplicated or dropped. The REST API response time for a concurrent GET /dashboard/kpis request during the burst is unaffected, confirming the decoupling of the ingestion layer from the REST layer.
Actual Result	All 100 events received HTTP 202 Accepted. Peak consumer group lag during the burst was 420 messages, well below the 10,000 threshold. The first event appeared in the Event Stream Feed after 0.4 seconds. All 100 events were confirmed present in the feed with no duplicates. A concurrent GET /dashboard/kpis request during the burst returned in 210 milliseconds, confirming that the Kafka ingestion had no impact on REST API latency.
Status	**Passed	100/100 events accepted, max lag = 420 messages, feed delay = 0.4s**

6. AI Output & Boundary Testing

This section validates that the Z AI GLM integration produces structurally acceptable outputs for the documented use cases, handles abnormal or oversized inputs gracefully, and that the system's hallucination detection mechanism activates correctly when the model returns an invalid response. All prompt/response pairs documented here are based on the actual inference calls made against the GLM during development.

6.1. Prompt/Response Test Pairs

A minimum of three prompt/response test pairs are documented below. For each pair, the acceptance criteria defines precisely what a correct, acceptable, and failing response looks like in the context of the FranchiseIQ use case.

ID	Prompt Input	Acceptance Criteria	Actual Output	Status	Notes
AI-01	POST /ai/pricing/optimize/store-kl-001 with weights: costSaving 80%, brandPresence 10%, revenueMaximization 10%	The recommendations list must contain at least one SKU where recommendedPrice is less than or equal to currentPrice, correctly reflecting the cost-saving priority. The projectedRevenue uplift must not be the highest possible value for this store.	All 6 SKUs returned recommendedPrice values at or below currentPrice. Brown Sugar Boba dropped from RM 8.90 to RM 8.50. projectedRevenue uplift was 3.2%, the lowest of any strategy combination tested.	**Passed**	GLM correctly interpreted cost-saving priority as a downward price signal.
AI-02	POST /ai/sentiment/analyze/store-kl-001 with a reviews dataset that contains 4 mentions of drinks being warm and one direct mention of freezer temperature.	The response must include at least one entry in the alerts array with severity of critical and a topic of Food Safety. The recommendedAction must reference an equipment inspection. The overallScore must be below 4.0 due to the negative cluster.	The response contained one critical alert with the topic Food Safety and the action Immediate equipment inspection of freezer unit B. The overallScore was 3.7, reflecting the negative cluster weight.	**Passed**	Temporal clustering correctly identified the 4-review pattern as an anomaly rather than isolated feedback.
AI-03	POST /ai/inventory/forecast/store-kl-001 with includeWeather true, includeSocial true, and a 7-day horizon during a simulated school holiday period.	The predictions list must include ICE-CREAM-VANILLA with a stockoutRisk of HIGH and a daysUntilStockout value of 3 or fewer. The explanation must reference the school holiday	ICE-CREAM-VANILLA was flagged as HIGH risk with daysUntilStockout of 2. The explanation included the school holiday calendar as a data variable with	**Passed**	School holiday signal was correctly weighted higher than social trends for an ice cream

ID	Prompt Input	Acceptance Criteria	Actual Output	Status	Notes
		calendar as one of the data variables with a non-zero weight.	weight 0.30, and the weather forecast at 0.35.		product in a hot-weather period.
AI-04	POST /ai/pricing/optimize/store-kl-001 with a mocked GLM response injected that returns a plain-text paragraph instead of a JSON object.	The system must not return a 500 Internal Server Error. It must detect the schema violation, log the failure to ai_decision_log, retry with the strict schema correction prompt, and if the retry also fails, return the cached PostgreSQL result with a CACHED_RESULT flag.	The system correctly detected the non-JSON response, logged it to ai_decision_log with a fail status, retried once, and after the second simulated failure returned the cached result with the CACHED_RESULT flag present in the response body.	**Passed**	Fallback chain operated exactly as designed. No 500 error was surfaced to the frontend.

6.2. Oversized and Larger Input Test

This section defines the maximum input size the system accepts, and documents what occurs when that limit is exceeded during a real inference call.

Field	Detail
Maximum Input Size	30 days of sales history per store, capped at 100 transactions per SKU within that window. Estimated at approximately 8,000 to 12,000 tokens depending on SKU count.
Input Used During Testing	A synthetic dataset of 90 days of history was injected into the DTO flattening service, representing approximately 3x the standard limit with 42 active SKUs and a full transaction log.
Expected Behavior	The rolling window truncation logic activates and keeps only the most recent 100 transactions per SKU. The total token count is reduced to within the 128,000-token context limit. The GLM receives a structurally valid, trimmed prompt and returns a valid response. A truncation_applied flag is included in the response metadata.
Actual Behavior	The DTO flattening service correctly truncated the 90-day dataset to the most recent 30 days per SKU, reducing the token count from an estimated 34,000 to 11,200. The GLM returned a valid pricing recommendation. The truncation_applied flag was present in the response with a value of true. No data from beyond the 30-day window was included in the explanation's data variables.
Status	**Passed | Token count reduced from ~34,000 to ~11,200 via rolling window truncation.**

6.3. Adversarial and Edge Prompt Test

This test submits an ambiguous or malformed prompt to the GLM integration to validate that the system responds predictably, without crashing, returning sensitive information, or producing a hallucinated result that bypasses schema validation.

Field	Detail
Adversarial Prompt Injected	The storeId field in the pricing request was set to a 2,000-character string of random unicode characters, and the weights payload was replaced with a JSON object containing 50 unexpected keys with null values.
Expected Behavior	The Spring Boot controller must reject the request at the input validation layer before it reaches the Use Case layer or the GLM. The response must be HTTP 400 Bad Request with a message describing the invalid storeId format. The malformed weights object must trigger the weight validation error, not an unhandled NullPointerException.
Actual Behavior	The controller's @Valid annotation on the request DTO caught both violations simultaneously. The response was HTTP 400 with a structured error object listing the two violated constraints: storeId must match pattern [a-z0-9-]{3,50} and tradeoffWeights must contain exactly the three defined keys. No NullPointerException or 500 error was triggered. The GLM was never called.
Status	**Passed | Input validated at controller layer, GLM never reached.**

6.4. Hallucination Handling

This section describes in detail how FranchiseIQ detects and mitigates hallucinated responses from the Z AI GLM, and documents the results of the targeted hallucination test conducted during the preliminary round.

The primary detection mechanism is a two-step validation pipeline. In the first step, LangChain4j's Structured Output Parser attempts to deserialise the raw GLM response into the target Java DTO class. If the JSON is structurally invalid, missing required fields, or contains fields of an unexpected type, this step throws a parsing exception.

In the second step, even for responses that pass structural parsing, the backend applies a semantic validation layer. For pricing responses, the PriceValidator checks that no recommendedPrice deviates from currentPrice by more than 20%. For sentiment responses, the system validates that the overallScore falls within the range of 1.0 to 5.0 and that all topic scores are non-negative. For inventory responses, it validates that daysUntilStockout is a non-negative integer and that confidence scores fall between 0.0 and 1.0.

The temperature parameter for all GLM inference calls is set to 0.1, which is the lowest value that still allows the model to produce meaningful variation across different store contexts. This was chosen after testing at 0.0, which occasionally produced identical recommendations for stores with meaningfully different sales histories, and at 0.3, which introduced more variance than was useful for a financial recommendation tool.

When a hallucination is detected, whether through structural parsing failure or semantic bound violation, the following sequence is triggered. First, the failure is logged to the ai_decision_log table with a fail status, the raw GLM response text, and the specific validation error. Second, the system retries the inference call once using a corrected prompt that includes stricter instruction such as Respond ONLY with a JSON object matching this exact schema. Third, if the retry also fails, the system fetches the most recent valid result from the corresponding cache table in PostgreSQL and returns it with a CACHED_RESULT flag in the response body.

During the preliminary round testing, hallucination handling was triggered intentionally by injecting a mocked GLM client that returned plain-text prose on the first call and a syntactically valid but semantically invalid JSON on the second call, with a recommendedPrice of RM 9,999.00 for Brown Sugar Boba. The PriceValidator correctly flagged the RM 9,999.00 value as exceeding the 20% sanity bound, the result was rejected, and the cached recommendation from the previous valid inference was returned with the CACHED_RESULT flag. The entire sequence completed within 340 milliseconds, and the frontend displayed the cached result without any visible error state to the user.

6.5. Test Strategy & Plan Sign-Off

In accordance with Agile principles, the testing activities documented in this QATD are conducted iteratively throughout the development cycle, not exclusively at the end. Unit tests are written alongside feature development during Phase 1 and Phase 2. Integration tests are added at the end of each phase when two or more components are stable enough to test together. Regression tests run on every merge to main, providing continuous validation that new changes do not break previously passing features.

The team acknowledges that some test cases, particularly the AI output tests in Section 6, may have their actual results updated as the GLM model or prompt templates are refined during Phase 2 of development. Any such update must be recorded in the version history of this document, with the previous actual result preserved and the reason for the change clearly documented.

This document is considered the baseline test plan for the FranchiseIQ preliminary round submission, and it is signed off by both team members as accurately reflecting the testing state of the system as of the submission date.